

Capacitor Lab: Charging Through a Resistor

Capacitor Lab · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/em5.html>

Curriculum links: circuits · electricity

Learning intentions

- I can describe the shape of a capacitor's charging curve over time.
- I can relate the time constant $\tau = RC$ to how quickly a capacitor charges.
- I can apply $Q = CV$ to calculate the maximum charge stored.

Getting started

Open the Capacitor Lab. Set a battery voltage, resistance and plate geometry, then close the switch and turn on the animation to watch the capacitor charge.

Tasks

1. Run the simulation and sketch the shape of the voltage-vs-time charging curve. Is the initial rate of charging faster or slower than the rate near the end?
2. Calculate the time constant $\tau = RC$ for your resistance and capacitance. Use the step buttons to find the time it takes the capacitor to reach about 63% of its maximum voltage, and compare it to your calculated τ .

Working:

3. Reset the simulation, double the resistance, and repeat. Does the capacitor charge faster or slower this time? Does this match what you would predict from $\tau = RC$?
4. Calculate the maximum charge the capacitor can store using $Q = CV$, and compare it to the simulation's displayed value at full charge.

Working:

Extension (optional)

Capacitors are used in real circuits for timing and for smoothing out voltage fluctuations. Using what you know about the time constant, explain why a larger resistance or capacitance would be chosen for a slower timing circuit.

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